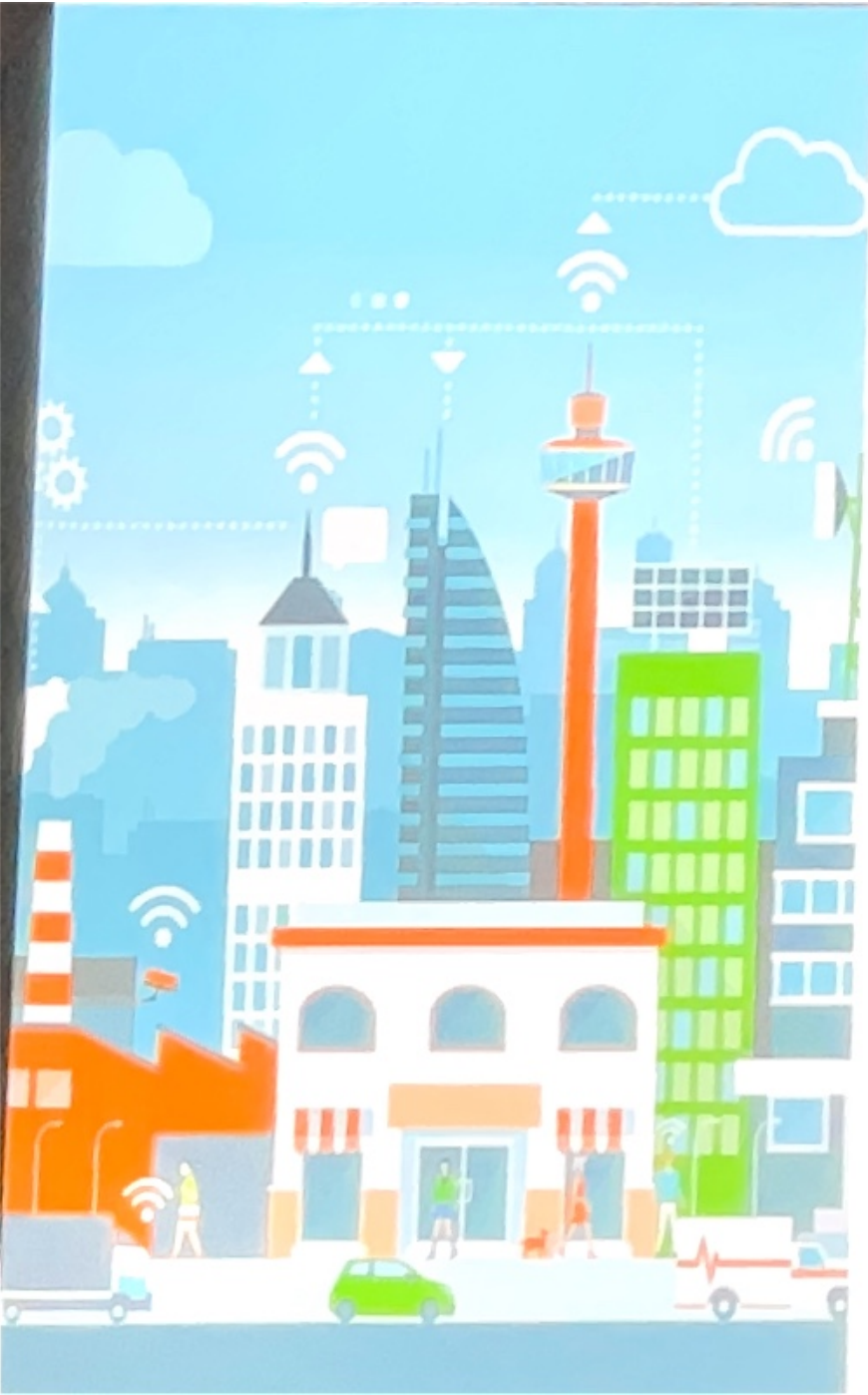




WHAT MAKES A CITY SMART?

INFOSYS 700 – Group 13



WHAT IS A SMART CITY?

- **Definition:** A smart city uses digital technologies and data to improve urban services, efficiency, and sustainability.
- **Core technologies:** IoT, AI, cloud computing, big data, 5G, blockchain.
- **Goal:** Enhance quality of life for residents and reduce environmental impact.



ROLE OF IOT IN SMART CITIES

- **Definition of IoT:** Network of connected physical devices that collect and exchange data.
- **How it helps:**
 - Real-time monitoring
 - Automation
 - Predictive maintenance
 - **Data-driven decisions**

DIGITAL IDENTITY MANAGEMENT

- Digital ID on blockchain stored in wearables or cards
- Real time authentication and verification
- Enhanced privacy protection by IoT devices verifying identity
- Seamless access to public services
- Easier reconciliation during a crisis
- SingPass – Singapore



IoT
DEVICES



DIGITAL
IDENTITY



CITY
SERVICES



CITY
SERVICES

ENERGY MANAGEMENT

- Smart grids - Bidirectional flow
- Energy data collected via IoT sensors in real time
- Demand-response systems to optimise usage
- Integration of renewable energy sources into the grid
- Personalised energy insights for consumers via smart meters
- Key Technologies: AMI, SCADA, DER



SMART TRANSPORTATION

- **Traffic Control:** Real-time traffic monitoring using IoT, cameras, and AI to reduce congestion
- Adaptive traffic lights, based on volume
- **Public Transport:** GPS-enabled transport offers real-time arrival updates via apps
- Digital ticketing and contactless payments for seamless access, use of cards or tap to pay.
- **Shared Vehicles:** Users locate, unlock, and pay for shared vehicles via app
- Sustainability focused,; hybrid and electric vehicles, reduced carbon emissions
- **Barcelona Initiatives:** Enhanced public transport, shared vehicles, traffic management, smart parking etc.

TRANSPORTATION IN A SMART CITY



TRAFFIC CONTROL
Real-time Traffic Monitoring and
Adaptive Traffic Lights



PUBLIC TRANSPORT
Real-time Tracking and Digital Ticketing



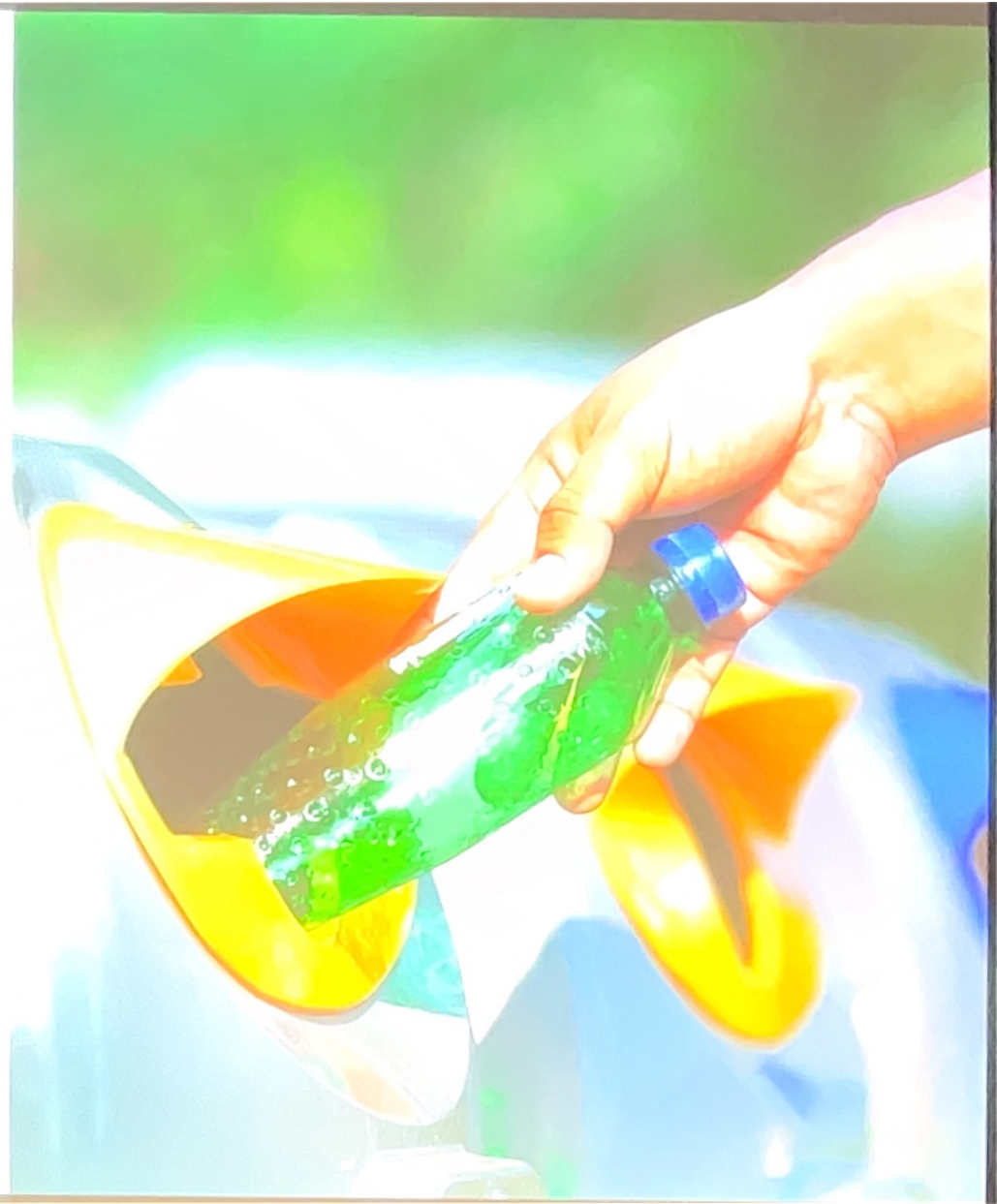
SHARED VEHICLES
App-Based Bikes, Scooters and Cars



SMART PARKING
Sensor-Enabled Parking and
Dynamic Pricing

WASTE MANAGEMENT

- IoT-enabled bins track fill levels in real time
- Route optimization reduces fuel use and CO₂ emissions
- Overflow alerts ensure timely collection and cleaner streets
- Improves resource allocation for local councils
- Supports sustainability and public hygiene goals
- **Barcelona:** Saved ~\$555,000/year through smart waste systems
- **Seoul:** Improved collection efficiency and reduced emissions
- **Market trend:** Global smart waste market to reach \$5.19B by 2025 (up from \$1.41B in 2018)



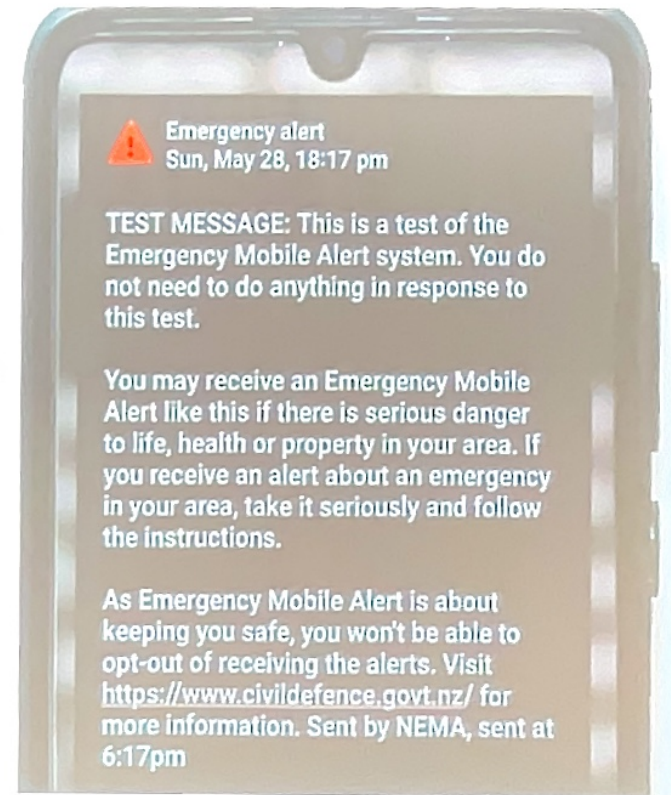
EMERGENCY MANAGEMENT

Emergency Alert System with IoT (NZ)

- Emergency Mobile Alert
- GeoNet
- Flood monitoring
- Fire/gas sensors
- Undersea cables

IoT Technology Involved

- **Sensors:** seismic, water level, smoke, gas, temperature
- **Networks:** 4G, LoRaWAN (low power, long range), satellite
- **Cloud platforms:** AWS, Azure – process & analyze data
- **Alerts:** mobile broadcasts, sirens, LED signs, emergency SMS



EMERGENCY MANAGEMENT

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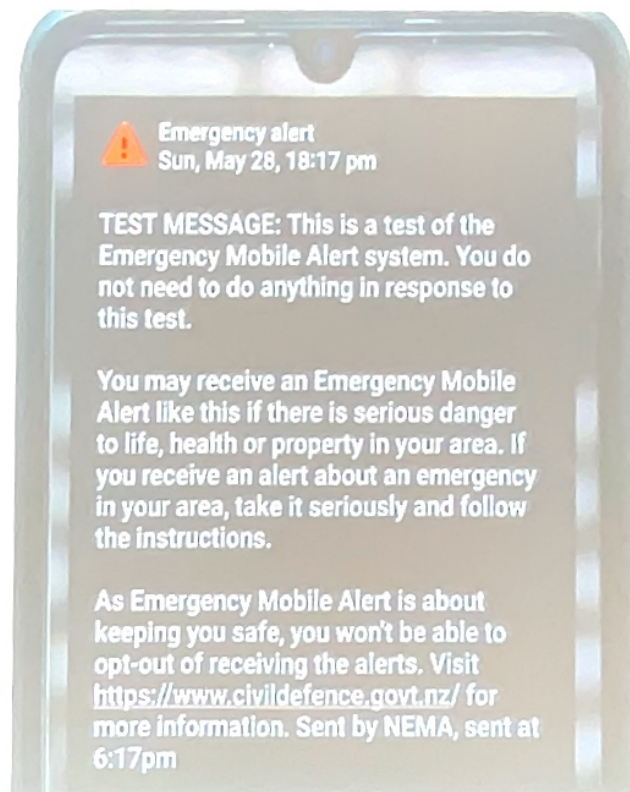
Benefits

- Early warnings save lives
- Real time automation reduces response delays
- Broad population reach

Challenges ?

IoT Technology Involved

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INFOSYS 700
Digital Innovation

SMART CITIES & IOT INTEGRATION

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